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AmicaIN / SmartIN / SMART2 - findings (full report)

Package:amica.com ·Version:1.0.50 ·Evidence: XAPK 2026-10-29, jadx, asset DATA.zip.

This document combines communication with the oven (reverse APK) and analysis of the Geryco/MKSoft ecosystem, support and verdict of the custom firmware.

Package:amica.com ·Badging version: 1.0.50 (compile/target SDK 36)

Stack manufacturer:MKSoft (https://github.com/search?q=mksoft+smart2) -SMART2 platform (white-label: Amica, Hansa, SmartIN).

Entry:com.mksoft.smart2.MainActivity ·Amica client ID: SMARTIN.ANDROID.LAN. (set in MainActivity).

> Note:folder FOTO/FIRMWARE in assets isOven screen wallpapers, not MCU files. Real OTA/USB described below.

1. Architecture (layers)

```
flowchart TB
subgraph ui [Declarative UI]
DATA[DATA.zip smartobj.xml + smarttxt]
USER[USER.zip recipes/photos]
end
subgraph core [R8-obfuscated core]
k3[k3.a application state singleton]
o3f[o3.f LAN/WAN/USB router]
end
subgraph lan [LAN / AP oven Wi-Fi]
udp[UDP discovery #SMT_AMC# / #PAE_AMC#]
tcp[TCP :7776 binary protocol q3.d]
end
subgraph wan [WAN relay]
https[HTTPS POST /AMICASMART/@timestamp]
p3[p3.a commands LOGUJ / SMART2_* / MEILING_*]
end
subgraph usb [USB flash drive]
amica[#AMICA#/DISO/*.cm2 + *.dat]
end
ui --> k3
k3 --> o3f
o3f -->|LAN| tcp
o3f -->|WAN| https
o3f -->|USB| amica
udp --> tcp
```

Layer | Class/file | Role

Global state | k3.a | Devices, Passwords, TCP Thread q3.e, Text Map, u() = Session ID

Network Config | u3.f | Ports, HTTP URLs, comm_mode, application_version

Mode router | o3.f | LAN → socket; WAN → p3.b + relay; USB → save files

TCP transfer | q3.d | Discovering the oven, status, upload files (firmware/recipe) in chunks 992 B

UDP broadcast | q3.a, q3.b | Searching for devices in LAN / configuration

JSON WiFi→oven | n3.c | SSID/password, fields M000, R001…, W001… (control)

HTTP client | p3.a | POST to 3 hosts, Base64 payload + o3.h cipher

USB export | r3.a | Catalog #AMICA# → DISO/

2. Connectivity Modes (comm_mode)

Enum in o3.f.b: LAN,WAN,USB (in UI equivalents AP/LAN/WAN from the Amica Home description - in this older app the nomenclature is in XML comm_mode).

Mode | Behavior (code) | User

LAN | Direct Socket to oven IP, port7776; refresh by q3.g/q3.f | Same WiFi as oven (or AP SmartIN)

WAN | Login LOGUJ/LOGUJ2, session CZY_AKTYWNA_SESJA, commands MEILING_DEVICE_* / SMART2_DEVICE_* via HTTPS | Remotely via the Amica/MKSoft server

USB | Saving .cm2/.dat packages on a #AMICA# pendrive or locally in USB mode | Updating/transferring data without TCP

Default configuration in u3.f (overwritten after GET_TEXT/FORMALNE from the server):

Parameter | Default value

tcp_port | 7776

udp_port | 7777

http | https://geryco.pl:7770/AMICASMART

http1 | https://g4help.com:7770/AMICASMART

http2 | https://g4help.pl:7770/AMICASMART

application_version (AmicaIN) | SMARTIN.ANDROID.LAN.

http_timeout / tcp_timeout | 1000 ms

refresh_oven | 1000 ms

HTTP:POST {base}/@{yyyyMMddHHmmss} - body = Base64(XOR/hash pipeline from o3.h).TLS:SSLSocketFactory withAllowAllHostnameVerifier (no pinning - MITM theoretically possible with bad CA).

3. Device discovery (UDP)

On startup/wizard, the application sends datagrams to broadcast (or udp_ip from config):

Magic | Use

#SMT_AMC# | Broadcast "I'm looking for an oven" (q3.a)

#PAE_AMC# / #PAG_AMC# | PAE/PAG configuration variant (q3.b, type f9951c / f9952d)

Payload: Session ID + phone IP (15 characters) + TCP port (6 characters) + counter - then Byte obfuscation (o3.h.d / o3.h.g with seed 123 or 238).

The oven response (TCP cmd 100, type 2) includes, among others: bytes ID / SOFT / HARD / NS - parsed in q3.d.d() as strings of the type SOFT:xx-yy, HARD:….

4. TCP protocol (control + file transfer)

Main logic: q3.d.w()- sending frames on the hostedq3.c (Socket).

Features (from decompilation):

- Message sequence number (16-bit LE) - must match the response.

- Chunked payload: max 992 bytes per frame on upload (z()).
- Auxiliary commands: including: 185(ping/init),296(transfer status),300 (Multiple Choices - transfer).
- Replies decrypted depending on device type (o3.h.d seed from f9962e).
- Separate thread q3.e maintains connection, handles timeouts, files id.bin / refridge_id.bin in the application directory.

4.1 Upload to oven (recipes, albums, firmware)

Common format (q3.d.y() / r3.a.e()):

1. Source file → transform o3.b.a() (compression/obfuscation).
2. Header: DZ + 3 fields + length + hash o3.h.c() based on session ID k3.a.u().
3. XML-like wrapper: <cm2>…hash…</cm2><data=…>…</data>(WAN uses thecm2 tag in p3.b.l).
4. TCP transfer in chunks; UI: TransferToOverActivity.

USB:the same packet saved as {id}.cm2 (header) + {id}.dat (payload) in/mnt/.../#AMICA#/DISO/ (r3.a).

Local USB mode (without pendrive): TransferToOverActivity writes .cm2 through k3.a.J().f() when comm_mode == USB.

Support files for recipe transfers: com.bin,tmp, extension.mtx.

5. Oven firmware - three paths

A) Update via server (WAN + version list)

1. UI: DATA/.../UPGRADE/, MAIN/AREA/UPGRADE, service menu (APP_UPG_* in smarttxt.xml).
2. Version metadata from the server parsed by t3.j→ listu3.g: soft_version, server_version, hard_version, soft_size,is_to_update, is_critical, device_id, folder, data (Base64).
3. Binary download: via the same HTTPS stack (p3.a.i / h) - response often Base64 decoded to local files.
4. After downloading: TCP upload(q3.d.t(u3.h) iterates and calls s())or prepare pendrive.

UX texts (PL): step 1 - put the pendrive in the oven and select update on the panel; step 2 - pendrive back to PC; etc. (UPGRADE/AREA/STEP*).

B) Pendrive (offline)

- Structure: folder #AMICA#→DISO/→ pairsNNN.cm2+NNN.dat.
- Compatible with r3.a.g() export.
- The license text (ul_pl.txt) explicitly mentions updates to the device controller software.

C) Update application itself AmicaIN (not oven MCU)

- Long press the language tile → POBIERZ_LNG → tmp.zip → unpack structures DATA (LangKafelItm).
- System SMART2 in the texts: local backup, rollback, "updating local structures".

Log/UI Thread: TCP_UW= "Software Update Thread" - refers to theupdate process, not the separate MQTT protocol.

6. WAN - HTTPS command directory (extracted strings)

Command | Objective

FORMALNE | Bootstrap: download XML config (t3.h → u3.f)

GET_TEXT | UI Texts (smarttxt)

LOGUJ / LOGUJ2 | WAN registration/login

CZY_AKTYWNA_SESJA | Session validation

SMART2_POBIERZ_ID | Device/client ID

SMART2_DEVICE_STATE / SMART2_DEVICE_REQUEST | Status and requests in SMART2 mode

MEILING_DEVICE_STATE / MEILING_DEVICE_REQUEST | Status/requests after binding to target (MAC/oven id)

SMART2_POBIERZ_* | Cookbook, weather, chapters

POBIERZ_LNG | Language pack/update app

POBIERZ_LISTE_PRZEPISOW / POBIERZ_PRZEPIS_CMD | Recipes

KOMUNIKAT_LISTA_ROZDZIALOW | com.bin for cookbook

Responses: XML VFPwith <ret>OK</ret> or <ret>ERR</ret>; network error:NETERR.

7. JSON to oven (WiFi provisioning / status)

n3.c builds, among others:

```
{"target":"…","from":"…","type":N, … Wxxx/Rxxx/Mxxx/Exxx fields …}
```

and {"SSID":"…","password":"…"} - oven WiFi configuration (WiFiSettingActivity collects SSID/password).

Fields R001–R005,E000–E004,M000/M001 map oven states (temperatures, heating functions - n3.d, n3.e generate JSON fragments).

8. Android Permissions (Implications)

From badging.txt:

- INTERNET, ACCESS_WIFI_STATE, CHANGE_WIFI_STATE, ACCESS_FINE/COARSE_LOCATION (WiFi scan),
- LOCAL_MAC_ADDRESS,
- READ/WRITE_EXTERNAL_STORAGE (pendrive / files .cm2),
- DELETE_PACKAGES / REQUEST_DELETE_PACKAGES (legacy / updater? - requires runtime audit).

No BLUETOOTH_* - communication WiFi + TCP/UDP + HTTPS, not BLE.

9. AmicaIN vs Amica Home

| AmicaIN(amica.com) | Amica Home (amicahome.com)

Generation | SMART2 / SmartIN, MKSoft | Newer product (Play describes AP/LAN/WAN/AUTO)

This document | Yes v1.0.50 | Separate APK analysis

10. What still needs runtime verification

- Full TCP command code map (i7 in w(): 100, 185, 296, 300, …).
- Exact file format after o3.b.a() before uploading the firmware.
- Relay behavior geryco / g4helptoday (DNS/SSL cert) - hosts may be legacy; app has3 fallback URL.
- Passwords in USER_CONFIG.zip (zip encrypted - not unzipped).

11. Quick links in the project

- COMMUNICATION (this file) (OVEN-COMMUNICATION.md)
- badging (apk-static/2026-10-29__amicain/badging.txt)
- endpoints HTTPS (apk-static/2026-10-29__amicain/endpoints.txt)
- UPGRADE PL hint step 1 (assets-extract/2026-10-29/DATA/AMICA/PL/UPGRADE/AREA/STEP1/hint.txt)

Evidence: AmicaIN XAPK 2026-10-29, jadx → com.mksoft.smart2, k3/q3/p3/u3/r3, asset DATA.zip + strings classes.dex.

Part II - SmartIN / SMART2 (Deep)

Supplement OVEN-COMMUNICATION.md (OVEN-COMMUNICATION.md). Sources: reverse AmicaIN 1.0.50 + public documentation + forums.

1. Who is behind it (ecosystem map)

Subject | Role

Amica S.A. | Household appliances manufacturer, brand, service, newerAmica Home/Amica Smart (Tuya)

Geryco sp. z o. o. (Kraków / historically also UK GERYCO LTD) | "Smart" technical support, portal geryco.pl/Home/Smart (https://geryco.pl/Home/Smart), HTTPS hosts in apk (geryco.pl, g4help.com/pl)

MKSoft (com.mksoft.smart2 in APK) | SMART2 platform - application engine, TCP/UDP protocol, .cm2 format, AMICASMART backend

SmartIN | The oven Wi-Fi access point name (SSID) and product line; in the code: SMARTIN.ANDROID.LAN.*, packages amica.com, smartin.com, com.mksoft.smartin

White-label | Same engine: Hansa(HANSA.ANDROID.LAN.), genericSMART2.ANDROID.LAN.

Important: Internet searches for "SMART2" often lead to a TV dongle or Linux package manager, not an Amica oven. Search: mksoft smart2, SmartIN, geryco, AMICASMART.

2. Three different "generations" of Amica oven software

Don't get confused - this is three stacks:

Line | Equipment | App | MCU/panel update

Amica .IN + SmartIN WiFi | RJ45 + WiFi module, displayTiw | AmicaIN(amica.com) - SMART2 | MKSoft server → telephone → TCP/USB #AMICA#; old PC programAmica.IN.USB_DataTransfer (Geryco)

Amica Integra | LCD, oftenSD card in the programmer | Integra/USB_DataTransfer | IMG/clone SD, service pendrive; softPGxxx / MGxxx (Electrode)

Amica .Home / X-Type WiFi | Knob + WiFi, displayTc | Amica Home(amicahome.com) | OTAfrom the cloud (screenOtA), regulations: platformTuya IoT-different protocol than SMART2

AmicaIN ≠ Amica Home. SMART2 reverse engineering does not describe Tuya.

Amica Home Official Pairing Guide: Android Tc (https://www.amica.pl/media/wysiwyg/programatory/tc/android_manual.pdf) Pairing PDF.

3. How to operate it (SmartIN / SMART2) - practice

3.1 Oven Side (Geryco + APK)

Source: Geryco Amica (https://geryco.com.pl/Amica), geryco.pl Smart (https://geryco.pl/Home/Smart).

1. In the oven WiFi menu: AP=1,M=1,S=1 (WiFi enabled, remote control allowed).

2. WI mode (communication frame + OK): the first app connection registers the app ID in the oven "firewall".

3. Home router (LAN/WAN): 2.4 GHz, WPA2-PSK (TKIP/AES), ports7770, 7773, 7763open (TCP; in the apk it dominates locally7776/7777 - see reverse).

4. AP SmartIN: the phone connects directly to the oven AP (like a "remote control").

3.2 On the AmicaIN application side (reverse)

Step | Action

Installation | InstallActivity unpacks DATA.zip / USER.zip → working directory

Bootstrap WAN | FORMALNE + GET_TEXT → XML config (t3.h → ports, URLs)

Remote account | ClientActivity + LOGUJ / LOGUJ2 (requires previous LAN session according to messages in the apk)

Discovery | UDP #SMT_AMC# / #PAE_AMC# → TCP 7776

Control | JSON n3.c (temperatures, functions) + cyclic q3.d.r() (~76 B status)

Recipes/photos | Transfer DZ… + <cm2> + chunks 992 B or USB #AMICA#/DISO/

Oven firmware | List from server (t3.j, is_to_update) → HTTPS download → TCP/USB upload (see COMMUNICATION §5)

3.3 Windows (legacy, Geryco)

Amica.IN.USB_DataTransfer (Win10+) - recipes, albums, pendrive; does not replace full flashing of PG/MG Integra. Description on the Geryco website; users report thatAmica no longer supports the old file update ecosystem (Elektroda EB9333BA+ IN, 2024 (https://www.elektroda.pl/rtvforum/topic4057250.html)).

3.4 New OTA ovens (Amica Home / Tuya)

OTA marketing: over WiFi, message OtA, WiFi icon flashing (Icecat / Amica materials (https://objects.icecat.biz/objects/mmo_120046661_1749168948_653_541992.pdf)).No reverse engineering of Tuya: integration via the manufacturer's application, not via the 7770 SMART2 port.

4. SMART2 protocol - technical abbreviation (for integration)

Element | Value

AmicaIN customer ID | SMARTIN.ANDROID.LAN.{wersja}

TCP device | 7776 (default)

UDP | 7777

HTTPS relay | {geryco\ | g4help}:7770/AMICASMART POST /@{timestamp}

Payload WAN | Base64 + o3.h (XOR-like), no cert pinning

Upload | .cm2 + .dat, magic UDP/TCP #*_AMC#

Oven status | Not a full flash dump: SOFT/HARD version and a short status

Potential integration (Home Assistant etc.): LAN only, reverse-engineering q3.d and packet capture; no official API; WAN requires an MKSoft account and relay.

5. Custom Firmware/Community - Verdict

5.1 SmartIN / SMART2 / AmicaIN (WiFi MKSoft)

Question | Answer

Does anyone publish custom firmware for the SmartIN module? | Not Found - No repositories, exploits, SMART2 WiFi Oven "root".

GitHub mksoft / smart2 | Unrelated to Amica (e.g. mksoft-ltd (https://github.com/mksoft-ltd) - mobile applications; ubinux/smart2 (https://github.com/ubinux/smart2/) - package manager Linux).

DIY smart oven | Projects like SmartOven MQTT (https://github.com/banana-breads/SmartOven) use a custom oven and ESP32, not Amica SmartIN.

Conclusion: No community custom firmware for the MKSoft stack was found. Replacing an official .cm2 package or server response is a theoretical modification, still involving manufacturer software and a high risk of bricking the device.

5.2 Amica Integra / IN - soft PG/MG (programmer, SD, service pendrive)

This is another layer(panel/controller, e.g.PG351,MG351,PG353):

- Elektroda (https://poszukaj.elektroda.pl/szukaj,amica-piekarnik-oprogramowanie.html): users share official images.IMG / .7z to recover fromSD Card ErrormodeSerwis, pendrive -service recoveries, not "custom ROM".

- Example of sharing a batch: EBI81064WAA PG351.05 (https://www.elektroda.pl/rtvforum/topic4066841.html) (file from a forum member).

- EB9333BA+ IN: USB instructions exist, but Amica no longer provides the software or supports the update program (topic4057250 (https://www.elektroda.pl/rtvforum/topic4057250.html)).

No homebrew custom firmware was found, only factory images, clones, and service recovery material. Flashing an unmatched PG/MG image risks damaging the device.

5.3 Amica Smart / Home (Tuya)

Terms and conditions: application for Tuya IoT(PDF agreement (https://amica-group.de/media/wysiwyg/AmicaSmart/PL_-_Amica_Smart_-_Regulamin_oraz_umowa_u_ytkownika_-_final.pdf)). Custom = Tuya ecosystem (other devices), not a documented Amica OTA hack. Factory OTA comes from Tuya/Amica servers.

6. Application firewall on the oven

Geryco states that the first connection in WI mode adds the application identifier to the oven whitelist. In the APK, session k3.a.u() (id.bin), the hash in LOGUJ, and MEILING_* fields with MAC/target indicate that the phone is not a generic TCP client without prior registration or known session secrets.

Implication: a custom LAN script must reproduce discovery and the handshake, or use an ID from a legitimately paired phone, only for the operator's own device.

7. Open directions (legal, own equipment)

1. Capture LAN (tcpdump on the router) during: discovery, heating on, firmware update - q3.d.w(…, i7, …) opcode mapping.

2. Comparison of Amica HomeAPK (Tuya) vsAmicaIN - do the new models use SMART2 at all.

3. Finding the current URL Amica.IN.USB_DataTransfer on Geryco (links often without a direct href in HTML - just a description).

4. Home Assistant: wrapper only after stable LAN status reading (without WAN login emulation).

8. External links (reading order)

1. Geryco - Smart / Amica IN (https://geryco.pl/Home/Smart) - Ports, WI, M/S/AP, USB_DataTransfer
2. Geryco Amica (PL) (https://geryco.com.pl/Amica)
3. Amica Home - Pairing PDF (https://www.amica.pl/media/wysiwyg/programatory/tc/android_manual.pdf)
4. Elektroda - Amica soft oven (index) (https://poszukaj.elektroda.pl/szukaj,amica-piekarnik-oprogramowanie.html)
5. Elektroda - not supported EB9333BA+ IN (https://www.elektroda.pl/rtvforum/topic4057250.html)
6. Amica Smart - Tuya Agreement PDF (https://amica-group.de/media/wysiwyg/AmicaSmart/PL_-_Amica_Smart_-_Regulamin_oraz_umowa_u_ytkownika_-_final.pdf)

Evidence: AmicaIN APK 2026-10-29, public web 2026-10-29. No confirmed SmartIN custom firmware in public sources.